



Module 4: AWS Cloud Security

AWS Academy Cloud Foundations

Module overview

Topics

- AWS shared responsibility model
- AWS Identity and Access Management (IAM)
- Securing a new AWS account
- Securing accounts
- Securing data on AWS
- Working to ensure compliance

Activities

- AWS shared responsibility model activity

Demo

- Recorded demonstration of IAM

Lab

- Introduction to AWS IAM



**Knowledge
check**

Module objectives

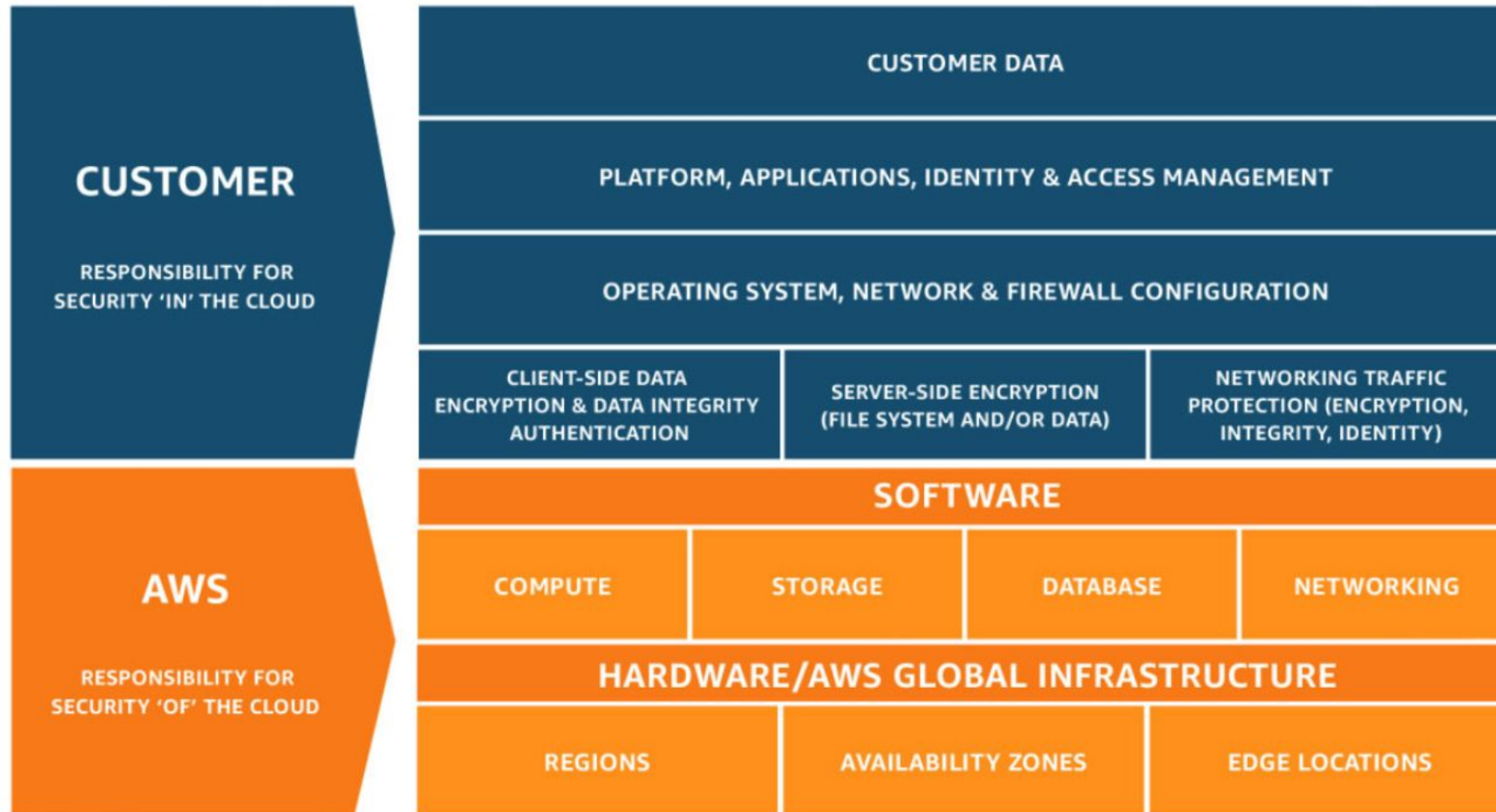
After completing this module, you should be able to:

- Recognize the shared responsibility model
- Identify the responsibility of the customer and AWS
- Recognize IAM users, groups, and roles
- Describe different types of security credentials in IAM
- Identify the steps to securing a new AWS account
- Explore IAM users and groups
- Recognize how to secure AWS data
- Recognize AWS compliance programs

Section 1: AWS shared responsibility model

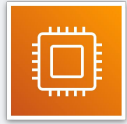
Module 4: AWS Cloud Security

AWS shared responsibility model



AWS responsibility: Security of the cloud

AWS services



Compute



Storage



Database



Networking

AWS Global
Infrastructure



Regions

Availability Zones



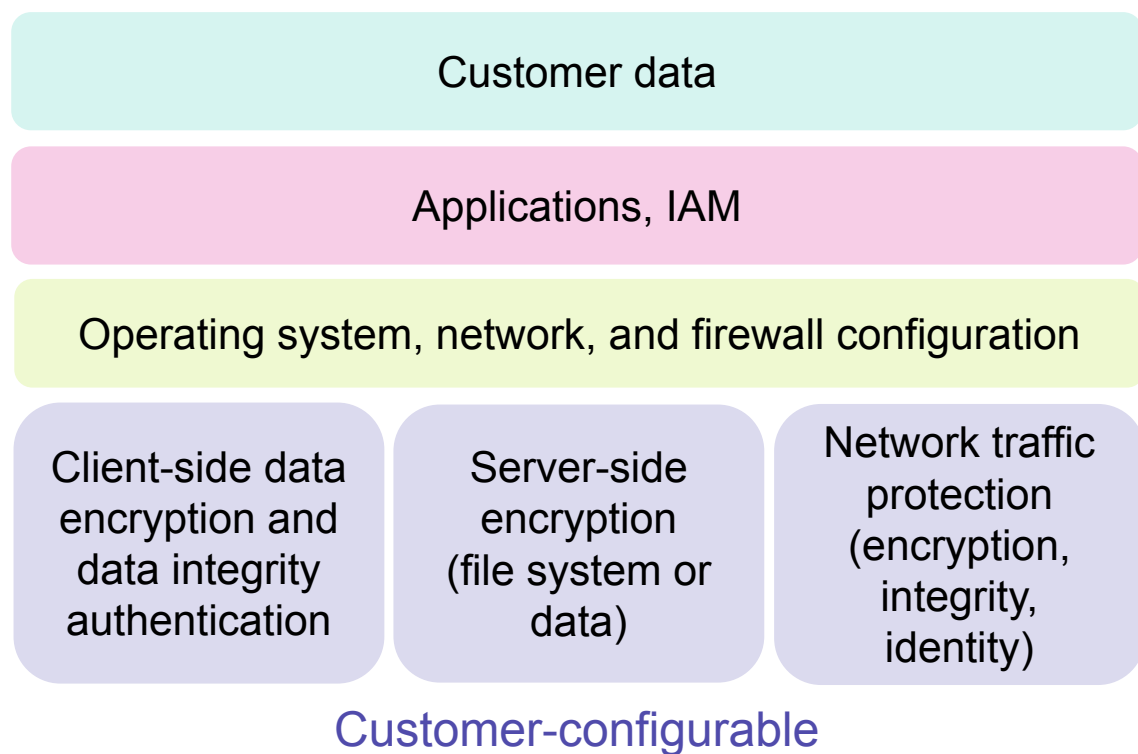
Edge locations

AWS responsibilities:

- Physical security of data centers
 - Controlled, need-based access
- Hardware and software infrastructure
 - Storage decommissioning, host operating system (OS) access logging, and auditing
- Network infrastructure
 - Intrusion detection
- Virtualization infrastructure
 - Instance isolation



Customer responsibility: Security *in* the cloud

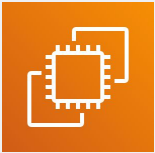


Customer responsibilities:

- Amazon Elastic Compute Cloud (Amazon EC2) instance **operating system**
 - Including patching, maintenance
- **Applications**
 - Passwords, role-based access, etc.
- **Security group** configuration
- OS or host-based **firewalls**
 - Including intrusion detection or prevention systems
- **Network** configurations
- Account management
 - Login and permission settings for each user

Service characteristics and security responsibility (1 of 2)

Example services managed by the customer



Amazon
EC2



Amazon Elastic
Block Store
(Amazon EBS)



Amazon
Virtual Private Cloud
(Amazon VPC)

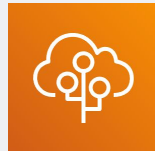
Example services managed by AWS



AWS
Lambda



Amazon
Relational Database
Service (Amazon
RDS)



AWS Elastic
Beanstalk

Infrastructure as a service (IaaS)

- Customer has more flexibility over configuring networking and storage settings
- Customer is responsible for managing more aspects of the security
- Customer configures the access controls

Platform as a service (PaaS)

- Customer does not need to manage the underlying infrastructure
- AWS handles the operating system, database patching, firewall configuration, and disaster recovery
- Customer can focus on managing code or data

Service characteristics and security responsibility (2 of 2)

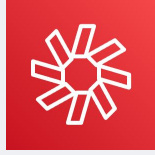
SaaS examples



AWS Trusted
Advisor



AWS Shield



Amazon Chime

Software as a service (SaaS)

- Software is centrally hosted
- Licensed on a subscription model or pay-as-you-go basis.
- Services are typically accessed via web browser, mobile app, or application programming interface (API)
- Customers do not need to manage the infrastructure that supports the service

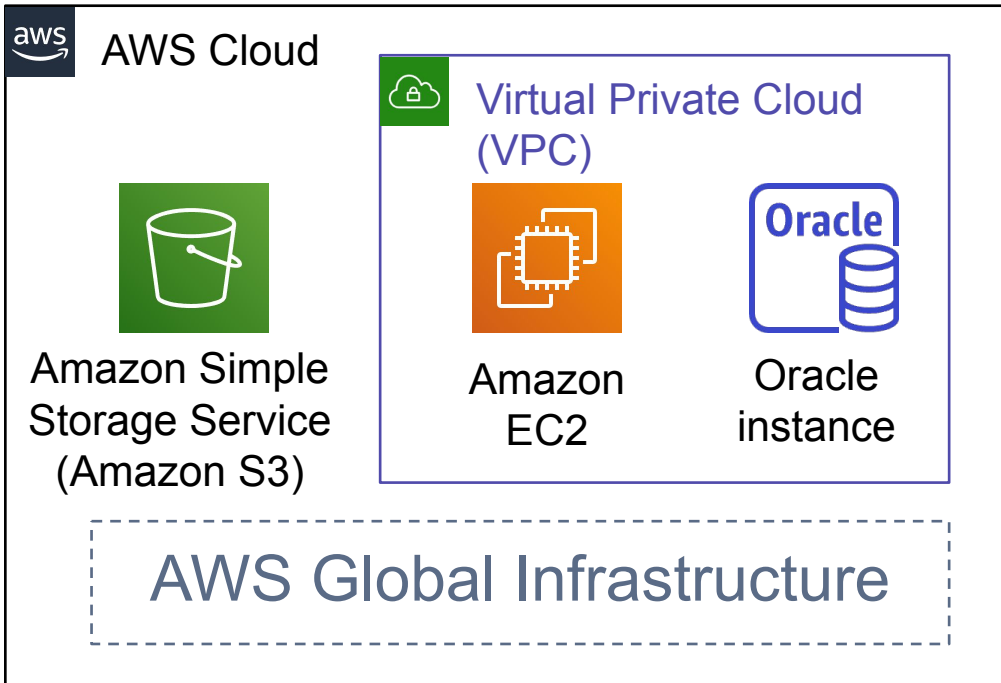
Activity: AWS shared responsibility model



Photo by Pixabay from Pexels.

Activity: Scenario 1 of 2

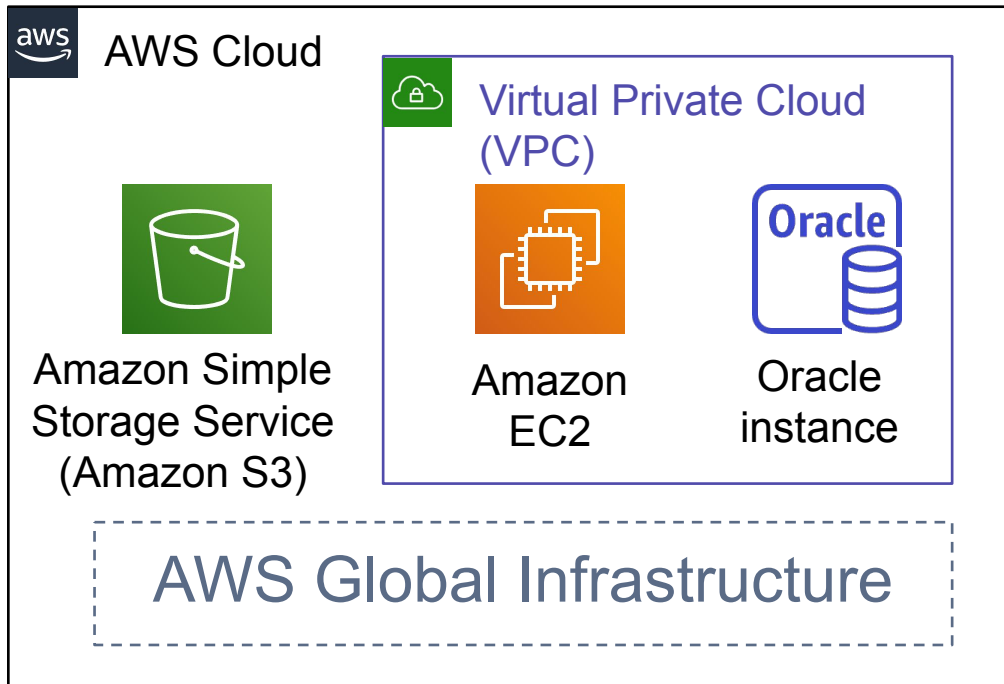
Consider this deployment. Who is responsible – AWS or the customer?



1. Upgrades and patches to the operating system on the EC2 instance?
2. Physical security of the data center?
3. Virtualization infrastructure?
4. EC2 security group settings?
5. Configuration of applications that run on the EC2 instance?
6. Oracle upgrades or patches If the Oracle instance runs as an Amazon RDS instance?
7. Oracle upgrades or patches If Oracle runs on an EC2 instance?
8. S3 bucket access configuration?

Activity: Scenario 1 of 2 Answers

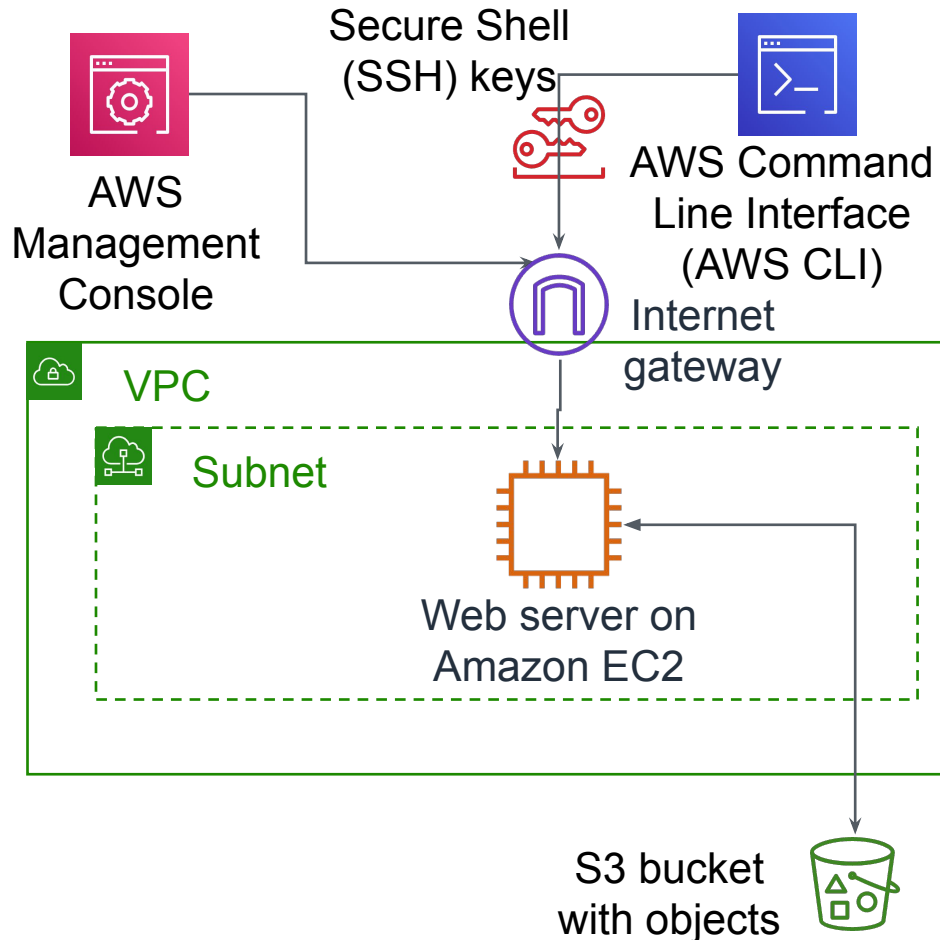
Consider this deployment. Who is responsible – AWS or the customer?



1. Upgrades and patches to the operating system on the EC2 instance?
 - **ANSWER:** The customer
2. Physical security of the data center?
 - **ANSWER:** AWS
3. Virtualization infrastructure?
 - **ANSWER:** AWS
4. EC2 security group settings?
 - **ANSWER:** The customer
5. Configuration of applications that run on the EC2 instance?
 - **ANSWER:** The customer
6. Oracle upgrades or patches If the Oracle instance runs as an Amazon RDS instance?
 - **ANSWER:** AWS
7. Oracle upgrades or patches If Oracle runs on an EC2 instance?
 - **ANSWER:** The customer
8. S3 bucket access configuration?
 - **ANSWER:** The customer

Activity: Scenario 2 of 2

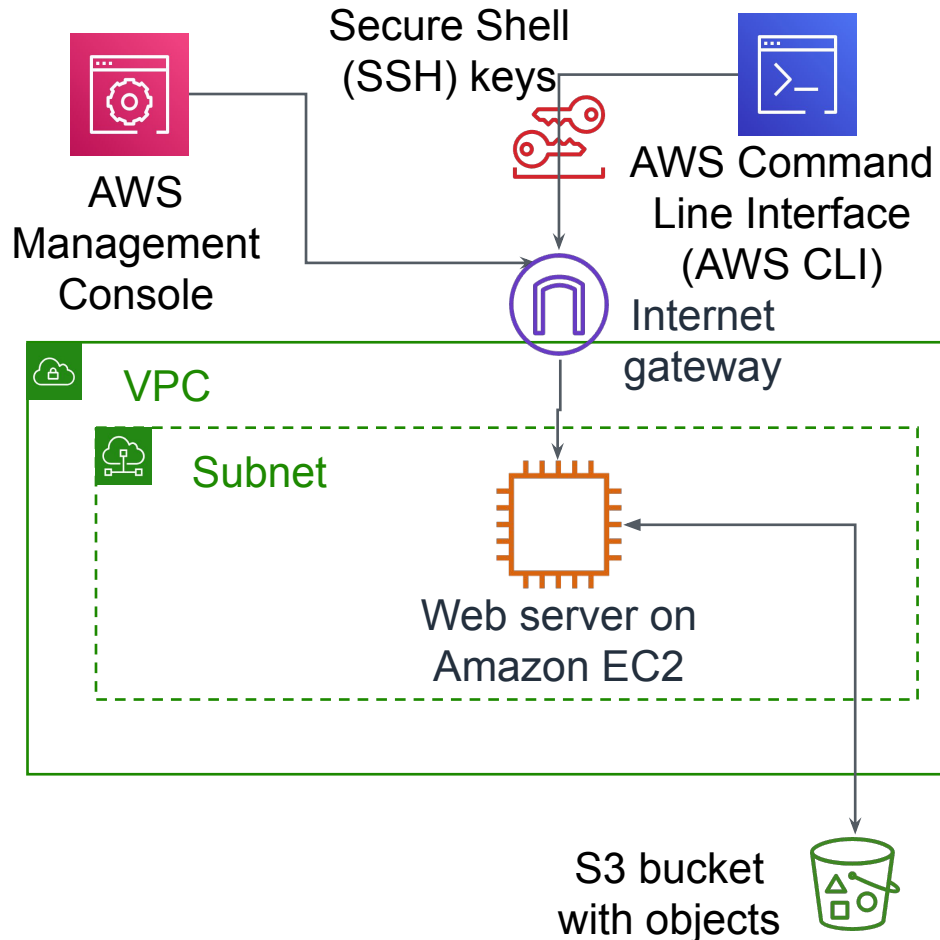
Consider this deployment. Who is responsible – AWS or the customer?



1. Ensuring that the AWS Management Console is not hacked?
2. Configuring the subnet?
3. Configuring the VPC?
4. Protecting against network outages in AWS Regions?
5. Securing the SSH keys
6. Ensuring network isolation between AWS customers' data?
7. Ensuring low-latency network connection between the web server and the S3 bucket?
8. Enforcing multi-factor authentication for all user logins?

Activity: Scenario 2 of 2 Answers

Consider this deployment. Who is responsible – AWS or the customer?



1. Ensuring that the AWS Management Console is not hacked?
• **ANSWER:** AWS
2. Configuring the subnet?
• **ANSWER:** The customer
3. Configuring the VPC?
• **ANSWER:** The customer
4. Protecting against network outages in AWS Regions?
• **ANSWER:** AWS
5. Securing the SSH keys
• **ANSWER:** The customer
6. Ensuring network isolation between AWS customers' data?
• **ANSWER:** AWS
7. Ensuring low-latency network connection between the web server and the S3 bucket?
• **ANSWER:** AWS
8. Enforcing multi-factor authentication for all user logins?
• **ANSWER:** The customer

Section 1 key takeaways



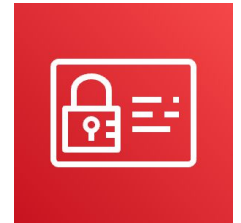
- AWS and the customer share security responsibilities:
 - AWS is responsible for security **of** the cloud
 - Customer is responsible for security **in** the cloud
- **AWS is responsible for protecting the infrastructure**—including hardware, software, networking, and facilities—that run AWS Cloud services
- For services that are categorized as infrastructure as a service (IaaS), the **customer is responsible for performing necessary security configuration and management tasks**
 - For example, guest OS updates and security patches, firewall, security group configurations

Section 2: AWS Identity and Access Management (IAM)

Module 4: AWS Cloud Security

AWS Identity and Access Management (IAM)

- Use **IAM** to manage access to **AWS resources** –
 - A resource is an entity in an AWS account that you can work with
 - Example resources; An Amazon EC2 instance or an Amazon S3 bucket
- *Example* – Control who can terminate Amazon EC2 instances
- Define fine-grained access rights –
 - **Who** can access the resource
 - **Which** resources can be accessed and what can the user do to the resource
 - **How** resources can be accessed
- IAM is a no-cost AWS account feature



AWS Identity and
Access Management
(IAM)

IAM: Essential components



IAM
user

A **person** or **application** that can authenticate with an AWS account.



IAM group

A **collection of IAM users** that are granted identical authorization.



IAM policy

The document that defines **which resources can be accessed** and the **level of access** to each resource.



IAM role

Useful mechanism to grant a set of permissions for making AWS service requests.

Authenticate as an IAM user to gain access

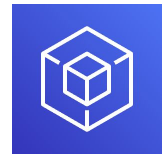
When you define an **IAM user**, you select what *types of access* the user is permitted to use.

Programmatic access

- Authenticate using:
 - Access key ID
 - Secret access key
- Provides AWS CLI and AWS SDK access



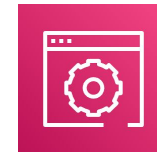
AWS CLI



AWS Tools
and SDKs

AWS Management Console access

- Authenticate using:
 - 12-digit Account ID or alias
 - IAM user name
 - IAM password
- If enabled, **multi-factor authentication (MFA)** prompts for an authentication code.



AWS Management
Console

IAM MFA

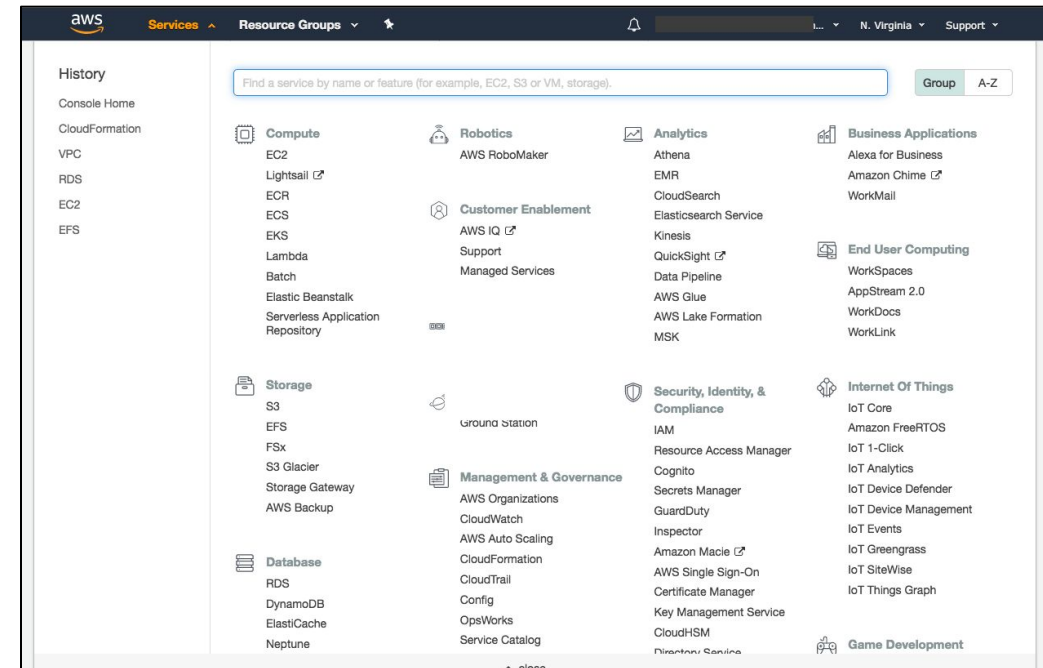
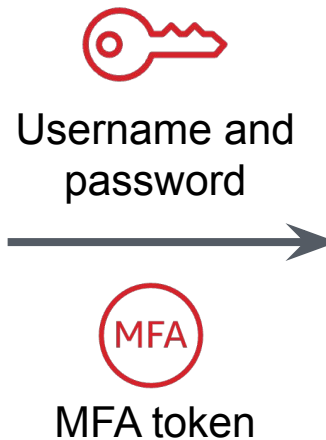
- MFA provides increased security.
- In addition to **username** and **password**, MFA requires a unique **authentication code** to access AWS services.

Account:

User Name:

Password:

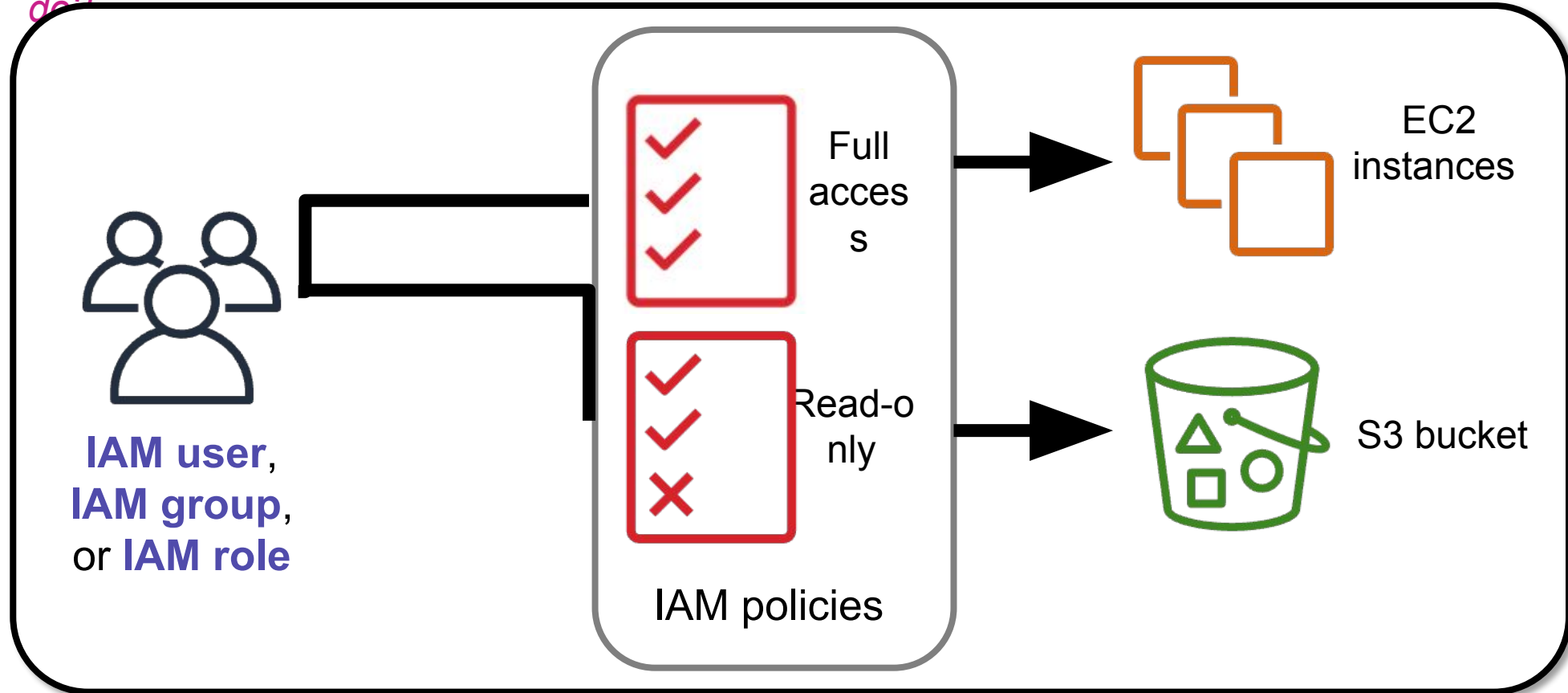
MFA users, enter your code on the next screen.



AWS Management Console

Authorization: What actions are permitted

After the user or application is connected to the AWS account, what are they allowed to do?



IAM: Authorization

- Assign permissions by creating an IAM policy.
- Permissions determine **which resources and operations** are allowed:
 - All permissions are implicitly denied by default.
 - If something is explicitly denied, it is never allowed.

Best practice: Follow the **principle of least privilege**.

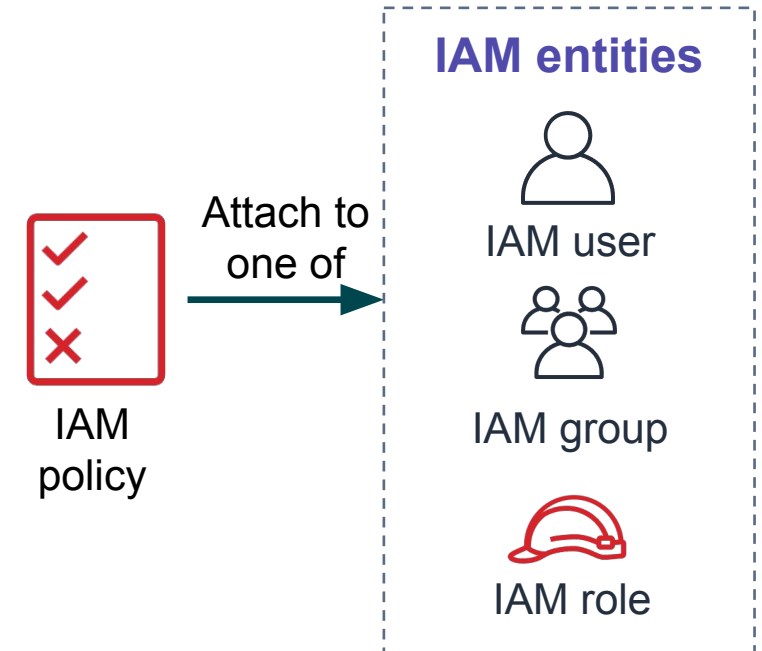


**IAM
permissions**

Note: The scope of IAM service configurations is **global**. Settings apply across all AWS Regions.

IAM policies

- **An IAM policy is a document that defines permissions**
 - Enables fine-grained access control
- Two types of policies – *identity-based* and *resource-based*
- **Identity-based** policies –
 - Attach a policy to any IAM entity
 - An **IAM user**, an **IAM group**, or an **IAM role**
 - Policies specify:
 - Actions that **may** be performed by the entity
 - Actions that **may not** be performed by the entity
 - A single *policy* can be attached to multiple *entities*
 - A single *entity* can have multiple *policies* attached to it
- **Resource-based** policies
 - Attached to a resource (such as an S3 bucket)



IAM policy example

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": ["DynamoDB:* ", "s3:*"],
    "Resource": [
      "arn:aws:dynamodb:region:account-number-without-hyphens:table/table-name",
      "arn:aws:s3:::bucket-name",
      "arn:aws:s3:::bucket-name/*"]
  },
  {
    "Effect": "Deny",
    "Action": ["dynamodb:*", "s3:*"],
    "NotResource": ["arn:aws:dynamodb:region:account-number-without-hyphens:table/table-name",
      "arn:aws:s3:::bucket-name",
      "arn:aws:s3:::bucket-name/*"]
  }
]
```

Explicit allow gives users access to a specific DynamoDB table and...

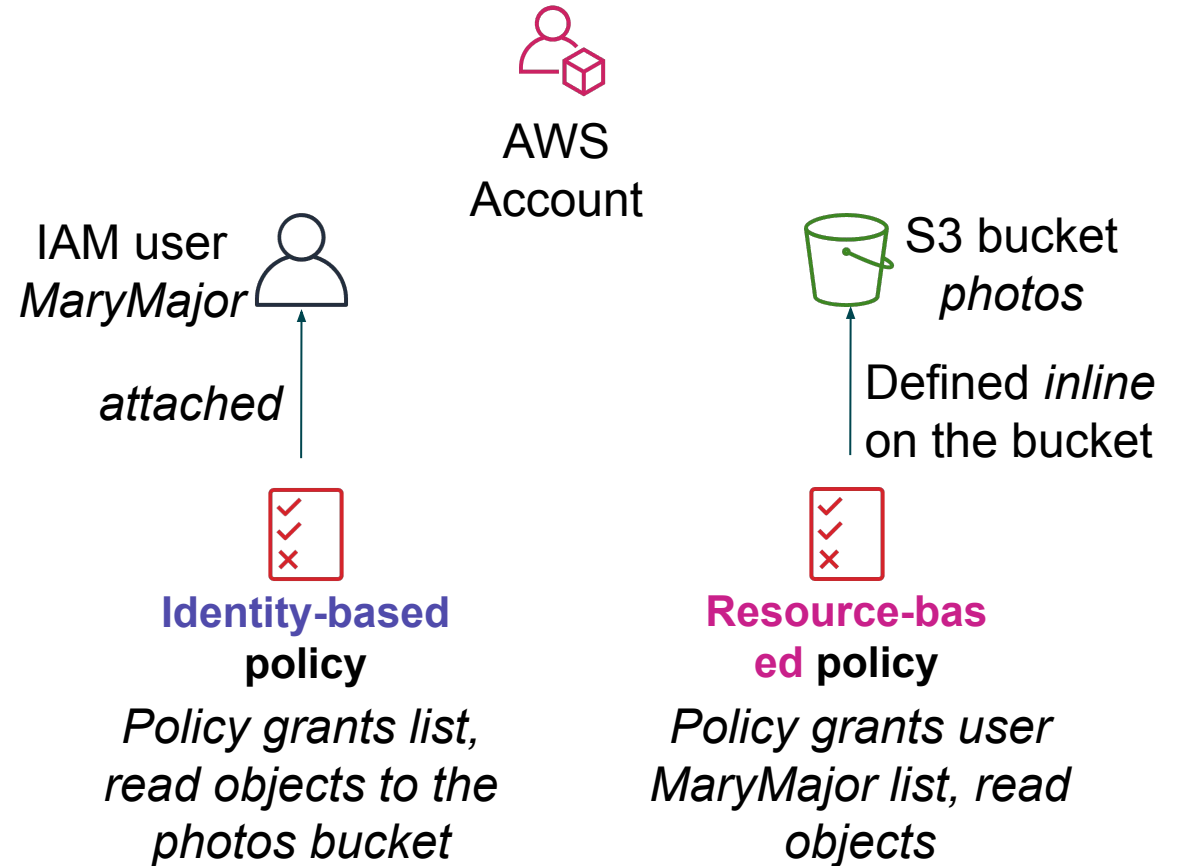
...Amazon S3 buckets.

Explicit deny ensures that the users cannot use any other AWS actions or resources other than that table and those buckets.

An explicit deny statement **takes precedence** over an allow statement.

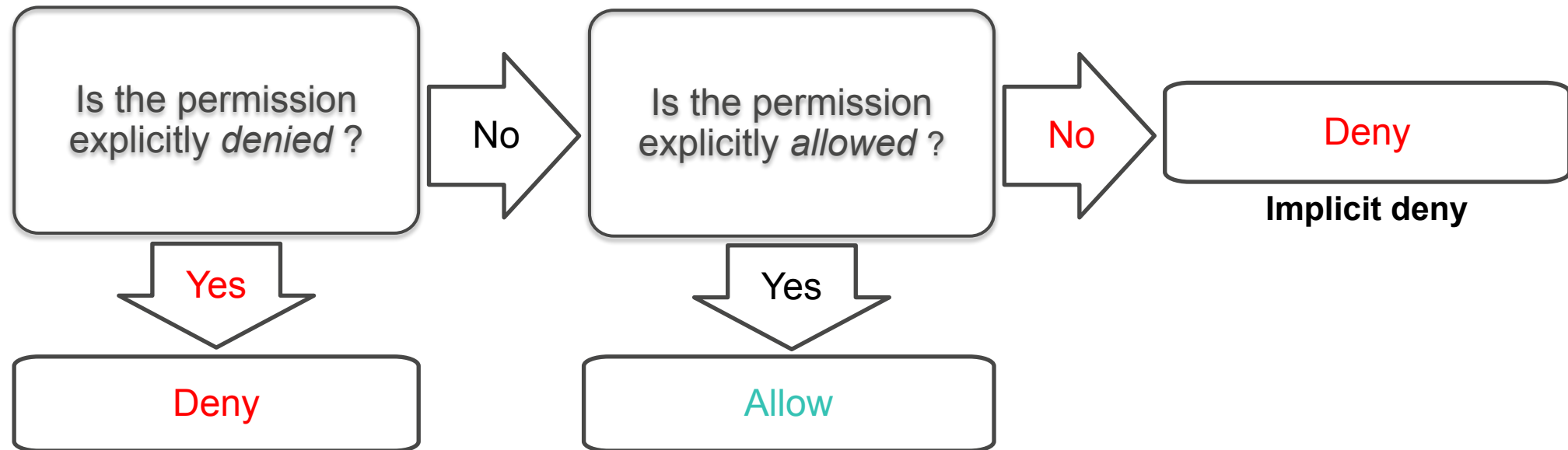
Resource-based policies

- *Identity-based policies* are attached to a user, group, or role
- **Resource-based policies** are attached to a resource (*not* to a user, group or role)
- Characteristics of resource-based policies –
 - Specifies who has access to the resource and what actions they can perform on it
 - The policies are *inline* only, not managed
- Resource-based policies are supported only by some AWS services



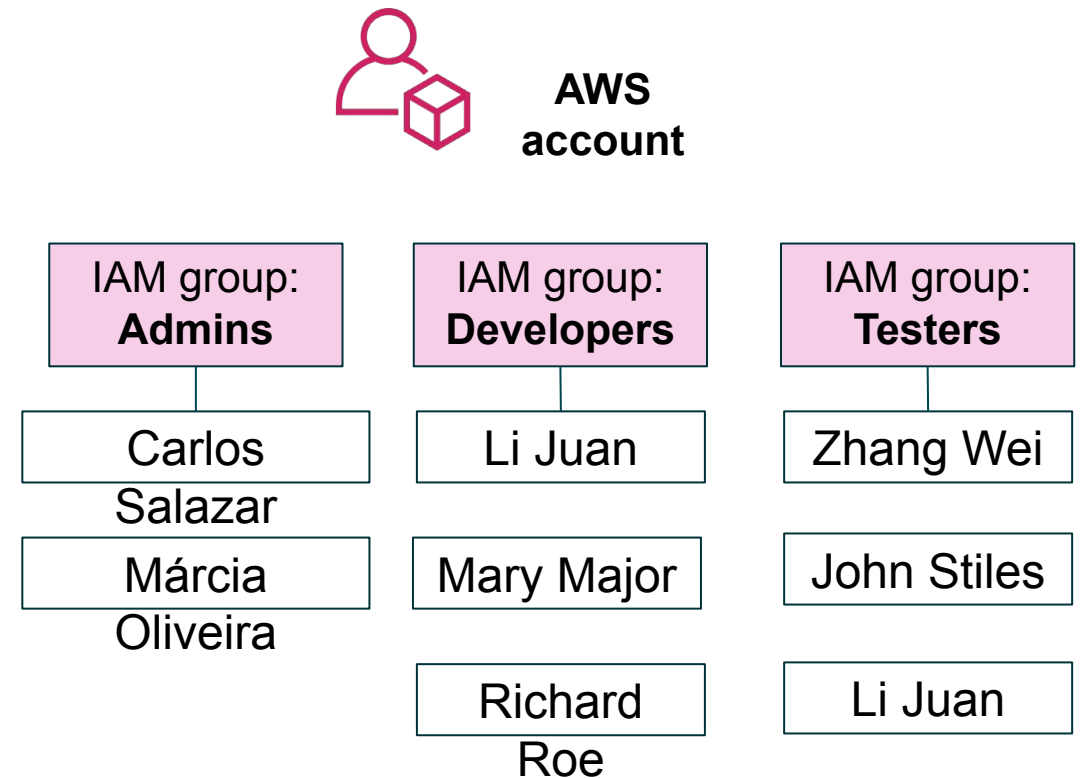
IAM permissions

How IAM determines permissions:



IAM groups

- An **IAM group** is a collection of IAM users
- A group is used to grant the same permissions to multiple users
 - Permissions granted by attaching IAM *policy* or policies to the group
- A user can belong to multiple groups
- There is no default group
- Groups cannot be nested



IAM roles

- An **IAM role** is an IAM identity with specific permissions
- Similar to an IAM user
 - Attach permissions policies to it
- Different from an IAM user
 - Not uniquely associated with one person
 - Intended to be *assumable* by a **person**, **application**, or **service**
- Role provides *temporary* security credentials
- Examples of how IAM roles are used to **delegate** access –
 - Used by an IAM user in the same AWS account as the role
 - Used by an AWS service—such as Amazon EC2—in the same account as the role
 - Used by an IAM user in a different AWS account than the role



IAM role

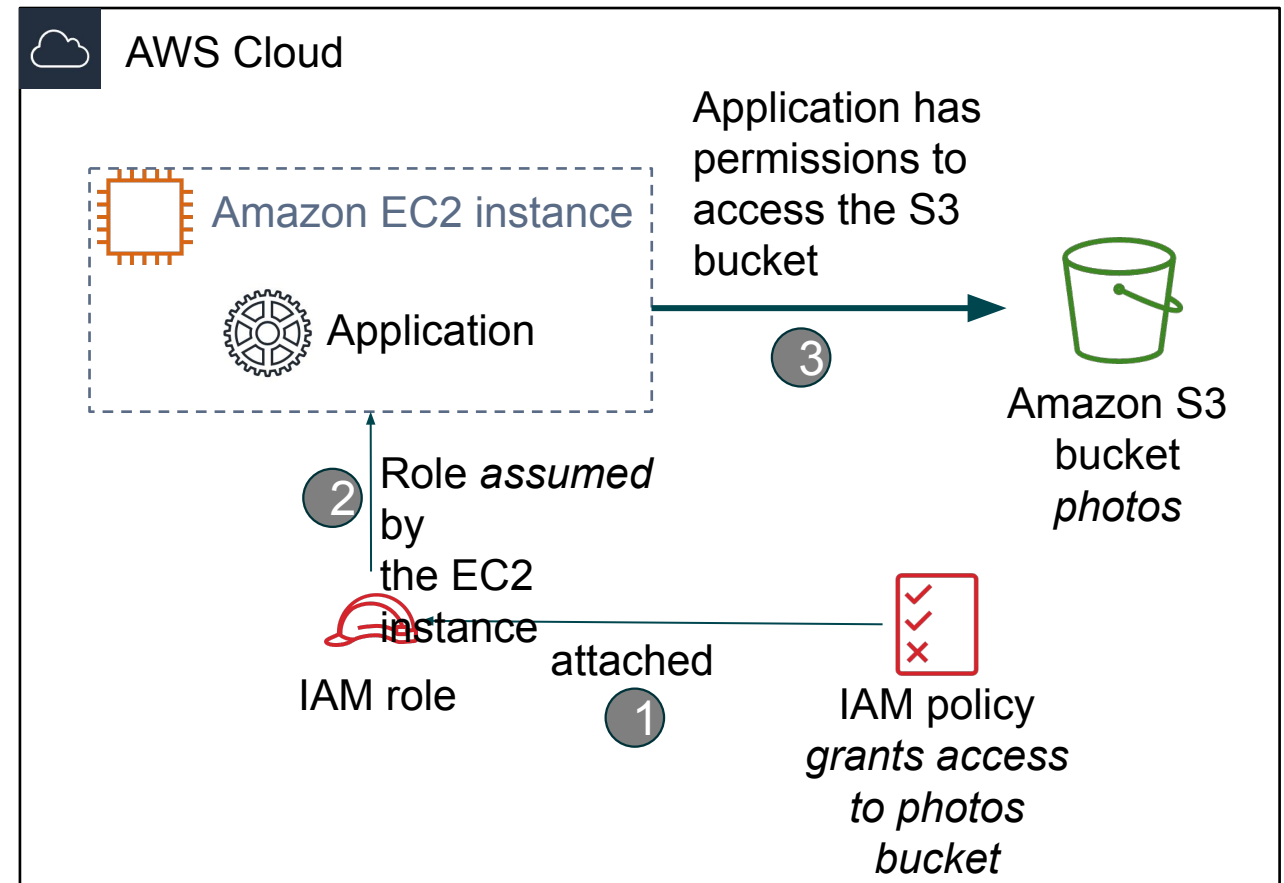
Example use of an IAM role

Scenario:

- An application that runs on an EC2 instance needs access to an S3 bucket

Solution:

- Define an IAM policy that grants access to the S3 bucket.
- Attach the policy to a role
- Allow the EC2 instance to assume the role

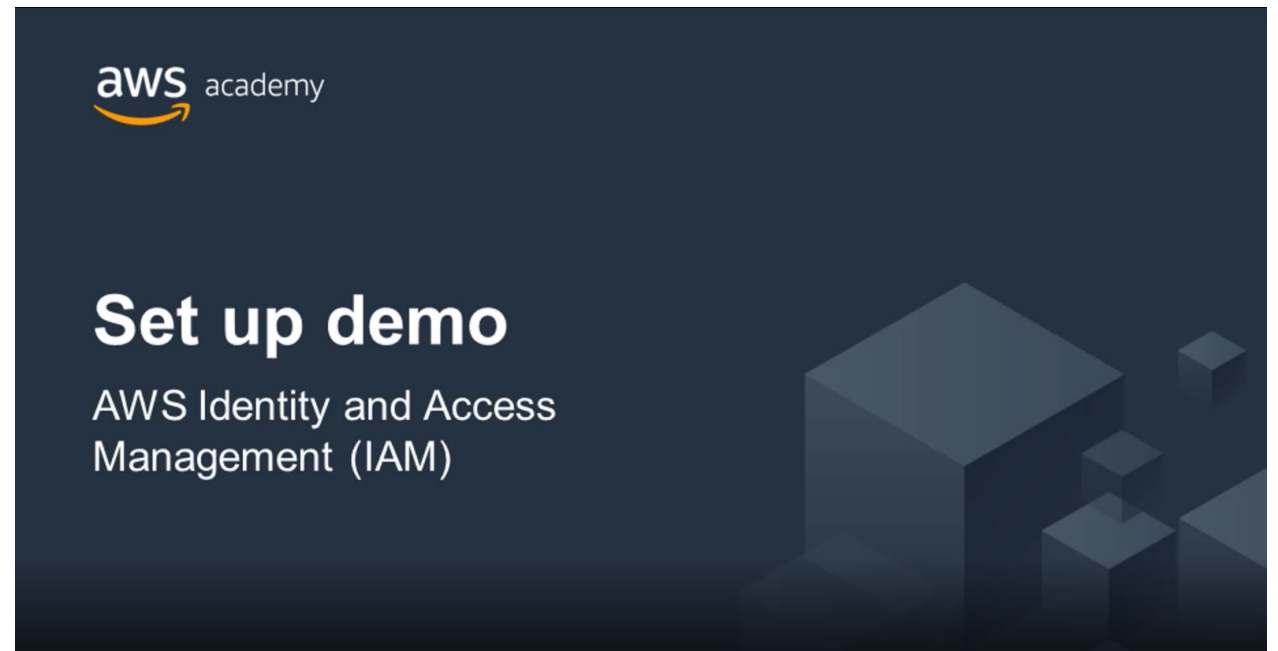


Section 2 key takeaways



- **IAM policies** are constructed with JavaScript Object Notation (JSON) and define permissions.
 - IAM policies can be attached to any **IAM entity**.
 - Entities are IAM users, IAM groups, and IAM roles.
- An **IAM user** provides a way for a person, application, or service to authenticate to AWS.
- An **IAM group** is a simple way to attach the same policies to multiple users.
- An **IAM role** can have permissions policies attached to it and can be used to delegate temporary access to users or applications.

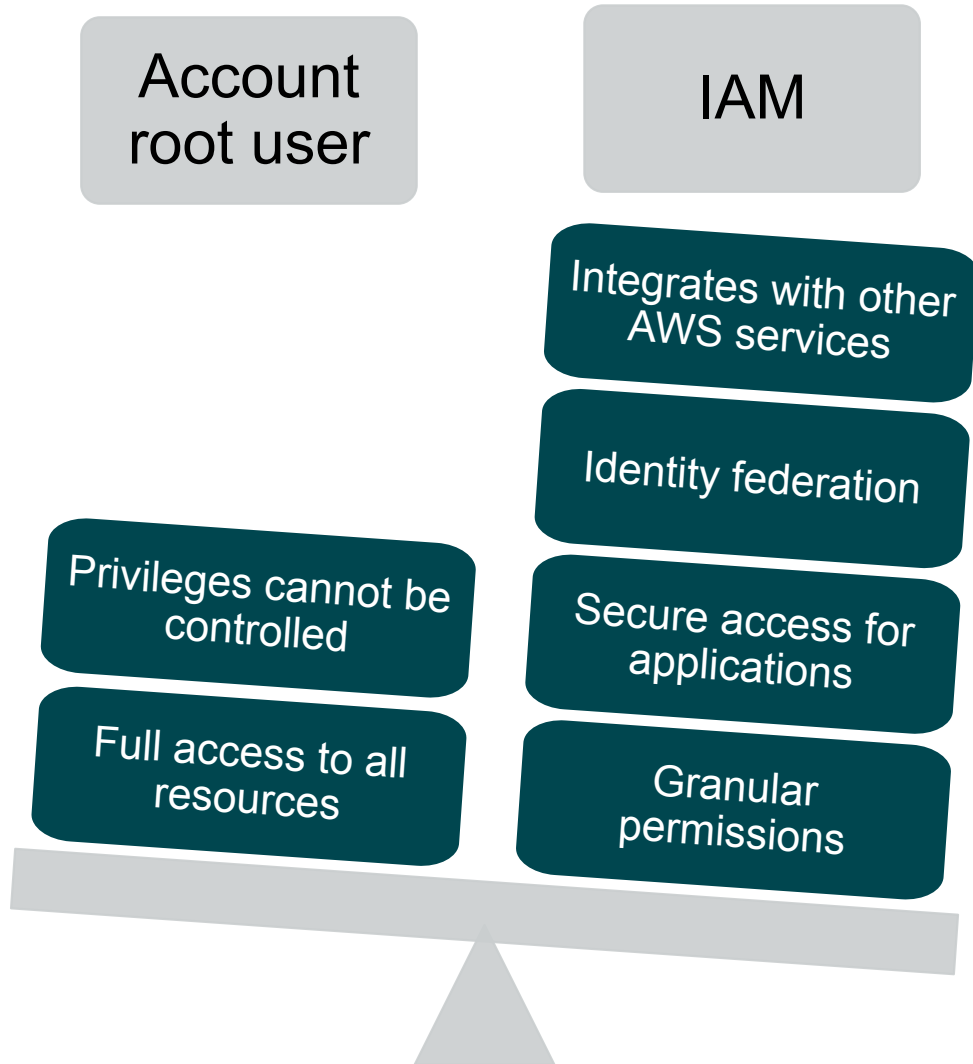
Recorded demo: IAM



Section 3: Securing a new AWS account

Module 4: AWS Cloud Security

AWS account root user access versus IAM access



- **Best practice:** Do not use the AWS account root user except when necessary.
 - Access to the **account root user** requires logging in with the *email address* (and password) that you used to create the account.
- Example actions that can only be done with the account root user:
 - Update the account root user password
 - Change the AWS Support plan
 - Restore an IAM user's permissions
 - Change account settings (for example, contact information, allowed Regions)

Securing a new AWS account: Account root user

Step 1: Stop using the account root user as soon as possible.

- The account root user has unrestricted access to all your resources.
- To stop using the account root user:
 1. While you are logged in as the account root user, **create an IAM user** for yourself. Save the access keys if needed.
 2. Create an IAM group, give it full administrator permissions, and add the IAM user to the group.
 3. Disable and **remove your account root user access keys**, if they exist.
 4. **Enable a password policy** for users.
 5. Sign in with your new IAM user credentials.
 6. Store your account root user credentials in a secure place.

Securing a new AWS account: MFA

Step 2: Enable multi-factor authentication (MFA).

- Require MFA for your [account root user](#) and for [all IAM users](#).
- You can also use MFA to control access to AWS service APIs.
- Options for retrieving the MFA token –
 - Virtual MFA-compliant applications:
 - Google Authenticator.
 - Authy Authenticator (Windows phone app).
 - U2F security key devices:
 - For example, YubiKey.
 - Hardware MFA options:
 - Key fob or display card offered by [Gemalto](#).



MFA token

Securing a new AWS account: AWS CloudTrail

Step 3: Use AWS CloudTrail.

- CloudTrail tracks user activity on your account.
 - Logs all API requests to resources in all supported services your account.
- Basic AWS CloudTrail event history is enabled by default and is free.
 - It contains all management event data on latest 90 days of account activity.
- To access CloudTrail –
 1. Log in to the **AWS Management Console** and choose the **CloudTrail** service.
 2. Click **Event history** to view, filter, and search the last 90 days of events.
- **To enable logs beyond 90 days and enable specified event alerting, create a trail.**
 1. From the CloudTrail Console trails page, click **Create trail**.
 2. Give it a name, apply it to all Regions, and create a new Amazon S3 bucket for log storage.
 3. Configure access restrictions on the S3 bucket (for example, only admin users should have access).

Securing a new AWS account: Billing reports

Step 4: Enable a billing report, such as the **AWS Cost and Usage Report**.

- Billing reports provide information about your use of AWS resources and estimated costs for that use.
- AWS delivers the reports to an Amazon S3 bucket that you specify.
 - Report is updated at least once per day.
- The **AWS Cost and Usage Report** tracks your AWS usage and provides estimated charges associated with your AWS account, either by the hour or by the day.

Optional: Securing a new AWS account – Full walkthrough

Module 4: AWS Cloud Security

IAM security status review

Search IAM

Custom Sign In Link

Welcome to Identity and Access Management

IAM users sign-in link:
https://.signin.aws.amazon.com/console [Customize](#) | [Copy Link](#)

IAM Resources

Users: 0 Roles: 0
Groups: 0 Identity Providers: 0
Customer Managed Policies: 0

Security Status 1 out of 5 complete.

- ✓ Delete your root access keys
- ⚠ Activate MFA on your root account
- ⚠ Create individual IAM users
- ⚠ Use groups to assign permissions
- ⚠ Apply an IAM password policy

Activate MFA on the account root user

Custom
sign-in
link

MFA
activation

Search IAM

Dashboard

Groups

Users

Roles

Policies

Identity providers

Account settings

Credential report

Encryption keys

Welcome to Identity and Access Management

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<https://console.signin.aws.amazon.com> [Customize](#) | [Copy Link](#)

IAM Resources

Users: 0 Roles: 0

Groups: 0 Identity Providers: 0

Customer Managed Policies: 0

Security Status

1 out of 5 complete.

- ✓ Delete your root access keys
- ⚠ Activate MFA on your root account
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Activate MFA on account root user

Search IAM

Dashboard

- Groups
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Credential reports

Encryption keys

Manage MFA device

If your virtual MFA application supports scanning QR codes, scan the following QR code with your smartphone's camera.



[Show secret key for manual configuration](#)

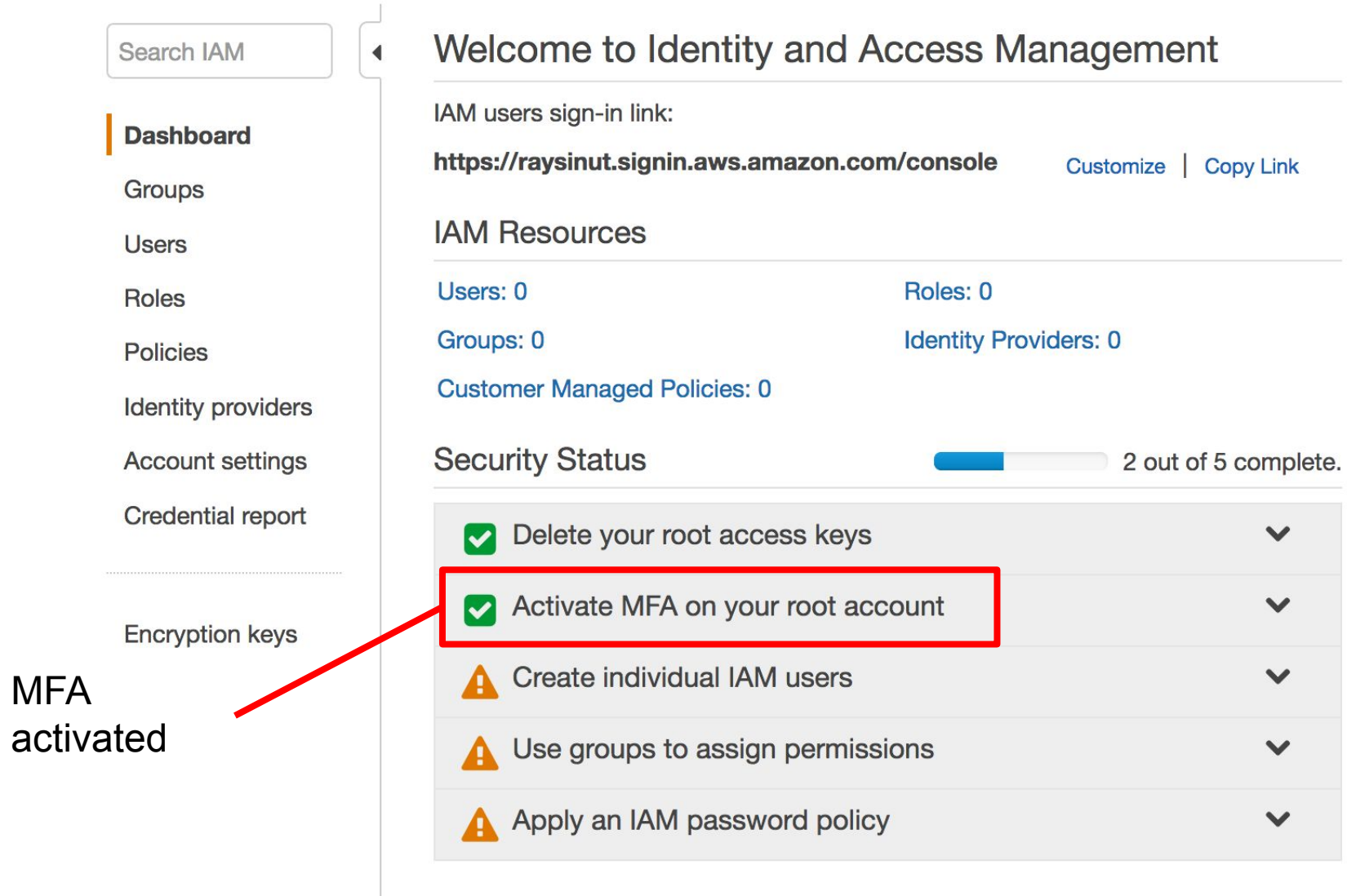
After the application is configured, enter two consecutive authentication codes in the boxes below and choose **Activate virtual MFA**.

Authentication code 1

Authentication code 2

[Cancel](#) [Previous](#) [Activate virtual MFA](#)

MFA on account root user is activated



The screenshot displays the AWS IAM console interface. On the left, a navigation sidebar includes a search bar labeled 'Search IAM' and a list of menu items: Dashboard, Groups, Users, Roles, Policies, Identity providers, Account settings, Credential report, and Encryption keys. The 'Dashboard' item is highlighted with an orange bar. The main content area is titled 'Welcome to Identity and Access Management'. It features a section for 'IAM users sign-in link' with the URL <https://raysinut.signin.aws.amazon.com/console> and links for 'Customize' and 'Copy Link'. Below this is the 'IAM Resources' section, showing counts for Users: 0, Roles: 0, Groups: 0, Identity Providers: 0, and Customer Managed Policies: 0. The 'Security Status' section includes a progress bar indicating '2 out of 5 complete'. A list of security tasks follows, each with a status icon and a dropdown arrow. The task 'Activate MFA on your root account' is highlighted with a red rectangular box and a red arrow pointing to it from the text 'MFA activated' on the left. Other tasks include 'Delete your root access keys' (checked), 'Create individual IAM users' (warning), 'Use groups to assign permissions' (warning), and 'Apply an IAM password policy' (warning).

Search IAM

Dashboard

Groups

Users

Roles

Policies

Identity providers

Account settings

Credential report

Encryption keys

Welcome to Identity and Access Management

IAM users sign-in link:

<https://raysinut.signin.aws.amazon.com/console> [Customize](#) | [Copy Link](#)

IAM Resources

Users: 0 Roles: 0

Groups: 0 Identity Providers: 0

Customer Managed Policies: 0

Security Status 2 out of 5 complete.

- ✓ Delete your root access keys
- ✓ **Activate MFA on your root account**
- ⚠ Create individual IAM users
- ⚠ Use groups to assign permissions
- ⚠ Apply an IAM password policy

MFA activated

Create an individual IAM user (1 of 5)

The screenshot displays the AWS IAM console interface. On the left is a navigation sidebar with a search bar labeled 'Search IAM' and a list of menu items: Dashboard (highlighted with an orange bar), Groups, Users, Roles, Policies, Identity providers, Account settings, Credential report, and Encryption keys. The main content area is titled 'Welcome to Identity and Access Management'. It includes a section for 'IAM users sign-in link' with the URL <https://raysinut.signin.aws.amazon.com/console> and links for 'Customize' and 'Copy Link'. Below this is the 'IAM Resources' section, showing counts for Users: 0, Roles: 0, Groups: 0, Identity Providers: 0, and Customer Managed Policies: 0. The 'Security Status' section features a progress bar indicating '2 out of 5 complete' and a list of recommendations. The first two recommendations, 'Delete your root access keys' and 'Activate MFA on your root account', are marked with green checkmarks. The third recommendation, 'Create individual IAM users', is marked with a yellow warning icon and is highlighted by a red rectangular box. A red arrow points from this box to the text 'IAM user creation' located to the left of the console. The remaining recommendations are 'Use groups to assign permissions' and 'Apply an IAM password policy', both marked with yellow warning icons.

Search IAM

Dashboard

Groups

Users

Roles

Policies

Identity providers

Account settings

Credential report

Encryption keys

Welcome to Identity and Access Management

IAM users sign-in link:

<https://raysinut.signin.aws.amazon.com/console> [Customize](#) | [Copy Link](#)

IAM Resources

Users: 0 Roles: 0

Groups: 0 Identity Providers: 0

Customer Managed Policies: 0

Security Status

2 out of 5 complete.

- ✓ Delete your root access keys
- ✓ Activate MFA on your root account
- ⚠ Create individual IAM users
- ⚠ Use groups to assign permissions
- ⚠ Apply an IAM password policy

IAM user creation

Create an individual IAM user (2 of 5)

Add user

1

Details

2

Permissions

3

Review

4

Complete

Set user details

You can add multiple users at once with the same access type and permissions. [Learn more](#)

User name*

Mic

[+ Add another user](#)

Select AWS access type

Select how these users will access AWS. Access keys and autogenerated passwords are provided in the last step. [Learn more](#)

Access type*



Programmatic access

Enables an **access key ID** and **secret access key** for the AWS API, CLI, SDK, and other development tools.



AWS Management Console access

Enables a **password** that allows users to sign-in to the AWS Management Console.

Console password*



Autogenerated password



Custom password

Require password reset



User must create a new password at next sign-in

Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

Create an individual IAM user (3 of 5)

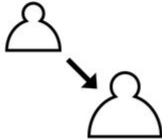
Add user



Set permissions for M



Add user to group



Copy permissions from existing user



Attach existing policies directly



Get started with groups

You haven't created any groups yet. Using groups is a best-practice way to manage users' permissions by job functions, AWS service access, or your custom permissions. Get started by creating a group. [Learn more](#)

Create group

Cancel

Previous

Next: Review

Create an individual IAM user (4 of 5)

Create group

Create a group and select the policies to be attached to the group. Using groups is a best-practice way to manage users' permissions by job functions, AWS service access, or your custom permissions. [Learn more](#)

Group name

Administrators

Create policy

Refresh

Filter: Policy type

Search

Showing 313 results

	Policy name	Type	Attachments	Description
☒	 AdministratorAccess	Job function	0	Provides full access to AWS services and resources.
☐	 AlexaForBusinessDeviceSetup	AWS managed	0	Provide device setup access to AlexaForBusiness services
☐	 AlexaForBusinessFullAccess	AWS managed	0	Grants full access to AlexaForBusiness resources and access to relat...
☐	 AlexaForBusinessGatewayEx...	AWS managed	0	Provide gateway execution access to AlexaForBusiness services
☐	 AlexaForBusinessReadOnlyA...	AWS managed	0	Provide read only access to AlexaForBusiness services
☐	 AmazonAPIGatewayAdminist...	AWS managed	0	Provides full access to create/edit/delete APIs in Amazon API Gatew...
☐	 AmazonAPIGatewayInvokeFu...	AWS managed	0	Provides full access to invoke APIs in Amazon API Gateway.
☐	 AmazonAPIGatewayPushToC...	AWS managed	0	Allows API Gateway to push logs to user's account.
☐	 AmazonAppStreamFullAccess	AWS managed	0	Provides full access to Amazon AppStream via the AWS Managemen...
☐	 AmazonAppStreamReadOnly...	AWS managed	0	Provides read only access to Amazon AppStream via the AWS Mana...

Cancel

Create group

Create an individual IAM user (5 of 5)

Add user

1

Details

2

Permissions

3

Review

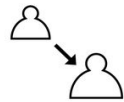
4

Complete

Set permissions for N



Add user to group



Copy permissions from
existing user



Attach existing policies
directly

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Create group

Refresh

Search

Showing 1 result

Group

Attached policies

☒ Administrators

AdministratorAccess

Cancel

Previous

Next: Review



IAM user creation successful

Add user



 **Success**

You successfully created the users shown below. You can view and download user security credentials. You can also email users instructions for signing in to the AWS Management Console. This is the last time these credentials will be available to download. However, you can create new credentials at any time.

Users with AWS Management Console access can sign-in at: <https://raysinut.signin.aws.amazon.com/console>

 Download .csv

	User	Access key ID	Secret access key	Password	Email login instructions
▶ 	Mi	AKI	***** Show	***** Show	Send email 

Close

IAM Dashboard security status

Search IAM

Dashboard

- Groups
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Credential report

Encryption keys

Password policy creation

Welcome to Identity and Access Management

IAM users sign-in link:
<https://raysinut.signin.aws.amazon.com/console> [Customize](#) | [Copy Link](#)

IAM Resources

Users: 1 Roles: 0
Groups: 1 Identity Providers: 0
Customer Managed Policies: 0

Security Status

4 out of 5 complete.

✓	Delete your root access keys	▼
✓	Activate MFA on your root account	▼
✓	Create individual IAM users	▼
✓	Use groups to assign permissions	▼
⚠	Apply an IAM password policy	▼

Set an IAM password policy

Search IAM

Dashboard

Groups

Users

Roles

Policies

Identity providers

Account settings

Credential report

Encryption keys

▼ Password Policy

You have unsaved changes to your password policy.

A password policy is a set of rules that define the type of password an IAM user can set. For more information about password policies, go to [Managing Passwords](#) in Using IAM.

Currently, this AWS account does not have a password policy. Specify a password policy below.

Minimum password length:

☒ Require at least one uppercase letter ⓘ

☒ Require at least one lowercase letter ⓘ

☒ Require at least one number ⓘ

☒ Require at least one non-alphanumeric character ⓘ

☒ Allow users to change their own password ⓘ

☐ Enable password expiration ⓘ
Password expiration period (in days):

☐ Prevent password reuse ⓘ
Number of passwords to remember:

☐ Password expiration requires administrator reset ⓘ

Apply password policy

Delete password policy

Security status checks completed

Search IAM

Dashboard

Groups

Users

Roles

Policies

Identity providers

Account settings

Credential report

Encryption keys

Welcome to Identity and Access Management

IAM users sign-in link:
<https://raysinut.signin.aws.amazon.com/console>
[Customize](#) | [Copy Link](#)

IAM Resources

Users: 1
Groups: 1
Customer Managed Policies: 0

Roles: 0
Identity Providers: 0

Security Status

5 out of 5 complete.

✓	Delete your root access keys	▼
✓	Activate MFA on your root account	▼
✓	Create individual IAM users	▼
✓	Use groups to assign permissions	▼
✓	Apply an IAM password policy	▼

Section 3 key takeaways



Best practices to secure an AWS account:

- **Secure** logins with multi-factor authentication (MFA).
- **Delete** account root user **access keys**.
- **Create** individual **IAM users** and grant permissions according to the principle of least privilege.
- **Use groups** to assign permissions to IAM users.
- **Configure** a **strong password policy**.
- **Delegate** using **roles** instead of sharing credentials.
- **Monitor** account activity by using AWS CloudTrail.

Lab 1: Introduction to IAM



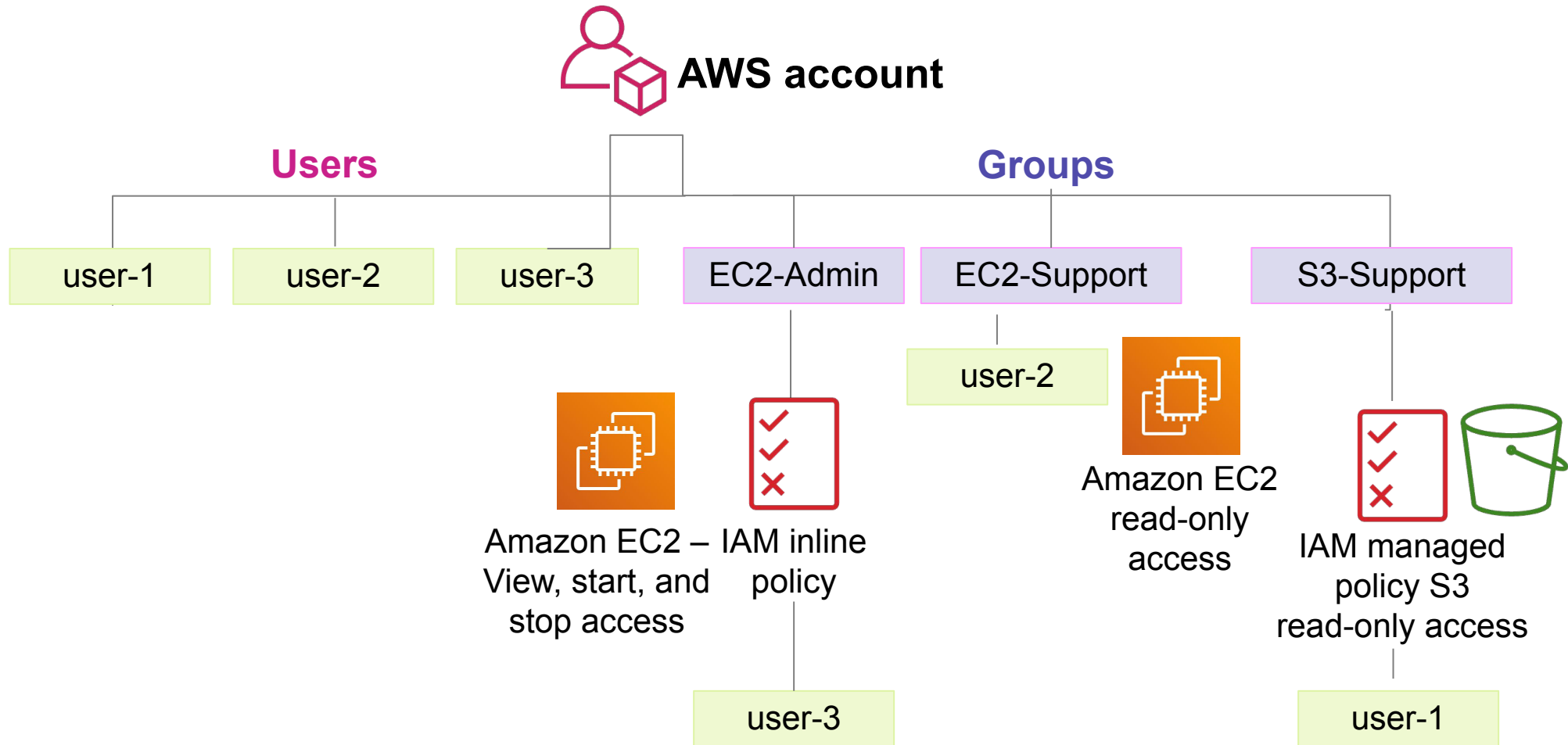
Lab 1: Tasks

- Task 1: Explore the Users and Groups.
- Task 2: Add Users to Groups.
- Task 3: Sign-In and Test Users.



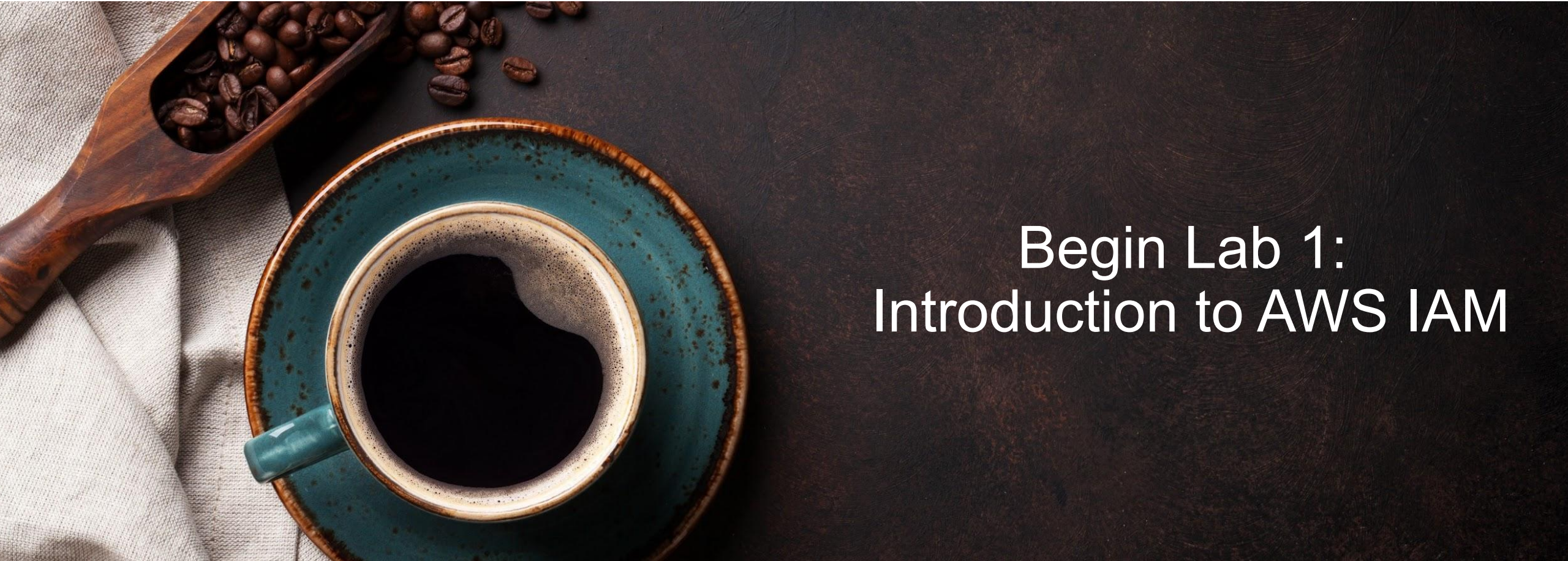
AWS Identity and
Access Management
(IAM)

Lab 1: Final product





~ 40 minutes



Begin Lab 1: Introduction to AWS IAM

Lab debrief: Key takeaways

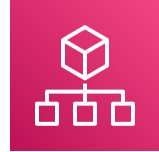


Section 4: Securing accounts

Module 4: AWS Cloud Security

AWS Organizations

- **AWS Organizations** enables you to consolidate multiple AWS accounts so that you centrally manage them.
- **Security features** of AWS Organizations:
 - **Group AWS accounts into organizational units** (OUs) and attach different access policies to each OU.
 - **Integration and support for IAM**
 - Permissions to a user are the intersection of what is allowed by AWS Organizations and what is granted by IAM in that account.
 - **Use service control policies** to establish control over the AWS services and API actions that each AWS account can access



AWS Organizations

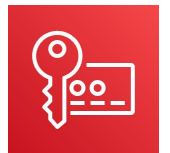
AWS Organizations: Service control policies

- **Service control policies (SCPs)** offer centralized control over accounts.
 - Limit permissions that are available in an account that is part of an organization.
- Ensures that accounts comply with access control guidelines.
- SCPs are *similar* to IAM permissions policies –
 - They use similar syntax.
 - However, an SCP never grants permissions.
 - Instead, SCPs **specify the maximum permissions** for an organization.

AWS Key Management Service (AWS KMS)

AWS Key Management Service (AWS KMS) features:

- Enables you to **create and manage encryption keys**
- Enables you to control the use of encryption across AWS services and in your applications.
- Integrates with AWS CloudTrail to log all key usage.
- Uses hardware security modules (HSMs) that are validated by Federal Information Processing Standards (FIPS) 140-2 to protect keys

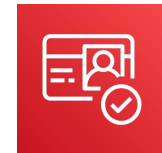


AWS Key Management
Service (AWS KMS)

Amazon Cognito

Amazon Cognito features:

- **Adds user sign-up, sign-in, and access control to your web and mobile applications.**
- Scales to millions of users.
- Supports sign-in with social identity providers, such as Facebook, Google, and Amazon; and enterprise identity providers, such as Microsoft Active Directory via Security Assertion Markup Language (SAML) 2.0.



Amazon Cognito

AWS Shield

- **AWS Shield** features:
 - Is a managed distributed denial of service (DDoS) protection service
 - Safeguards applications running on AWS
 - Provides always-on detection and automatic inline mitigations
 - *AWS Shield Standard* enabled for at no additional cost. *AWS Shield Advanced* is an optional paid service.
- Use it to **minimize application downtime and latency**.



AWS Shield

Section 5: Securing data on AWS

Module 4: AWS Cloud Security

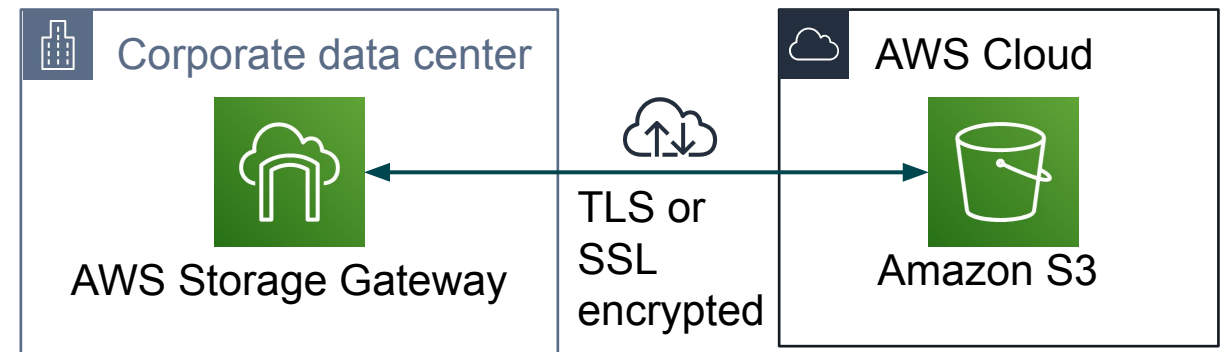
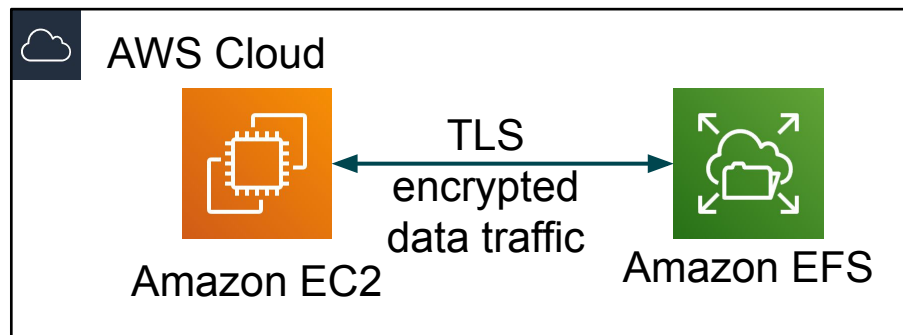
Encryption of data *at rest*

- **Encryption** encodes data with a **secret key**, which makes it unreadable
 - Only those who have the secret key can decode the data
 - **AWS KMS** can manage your secret keys
- AWS supports encryption of **data at rest**
 - Data at rest = Data stored physically (on disk or on tape)
 - You can encrypt data stored in any service that is supported by AWS KMS, including:
 - Amazon S3
 - Amazon EBS
 - Amazon Elastic File System (Amazon EFS)
 - Amazon RDS managed databases



Encryption of data *in transit*

- Encryption of **data in transit** (data moving across a network)
 - **Transport Layer Security (TLS)**—formerly SSL—is an open standard protocol
 - **AWS Certificate Manager** provides a way to manage, deploy, and renew TLS or SSL certificates
- Secure HTTP (HTTPS) creates a secure tunnel
 - Uses TLS or SSL for the bidirectional exchange of data
- **AWS services support data in transit encryption.**
 - Two examples:



Securing Amazon S3 buckets and objects

- Newly created S3 buckets and objects are **private** and **protected** by default.
- When use cases require sharing data objects on Amazon S3 –
 - It is essential to manage and control the data access.
 - Follow the **permissions that follow the principle of least privilege** and consider using Amazon S3 encryption.
- Tools and options for controlling access to S3 data include –
 - [Amazon S3 Block Public Access](#) feature: Simple to use.
 - IAM policies: A good option when the user can authenticate using IAM.
 - [Bucket policies](#)
 - [Access control lists](#) (ACLs): A legacy access control mechanism.
 - [AWS Trusted Advisor](#) bucket permission check: A free feature.

Section 6: Working to ensure compliance

Module 4: AWS Cloud Security

AWS compliance programs

- Customers are subject to many different security and compliance regulations and requirements.
- **AWS engages with certifying bodies and independent auditors to provide customers with detailed information about the policies, processes, and controls that are established and operated by AWS.**
- Compliance programs can be broadly categorized –
 - **Certifications and attestations**
 - Assessed by a third-party, independent auditor
 - Examples: [ISO 27001](#), [27017](#), [27018](#), and [ISO/IEC 9001](#)
 - **Laws, regulations, and privacy**
 - AWS provides security features and legal agreements to support compliance
 - Examples: EU [General Data Protection Regulation \(GDPR\)](#), HIPAA
 - **Alignments and frameworks**
 - Industry- or function-specific security or compliance requirements
 - Examples: Center for Internet Security (CIS), EU-US Privacy Shield certified

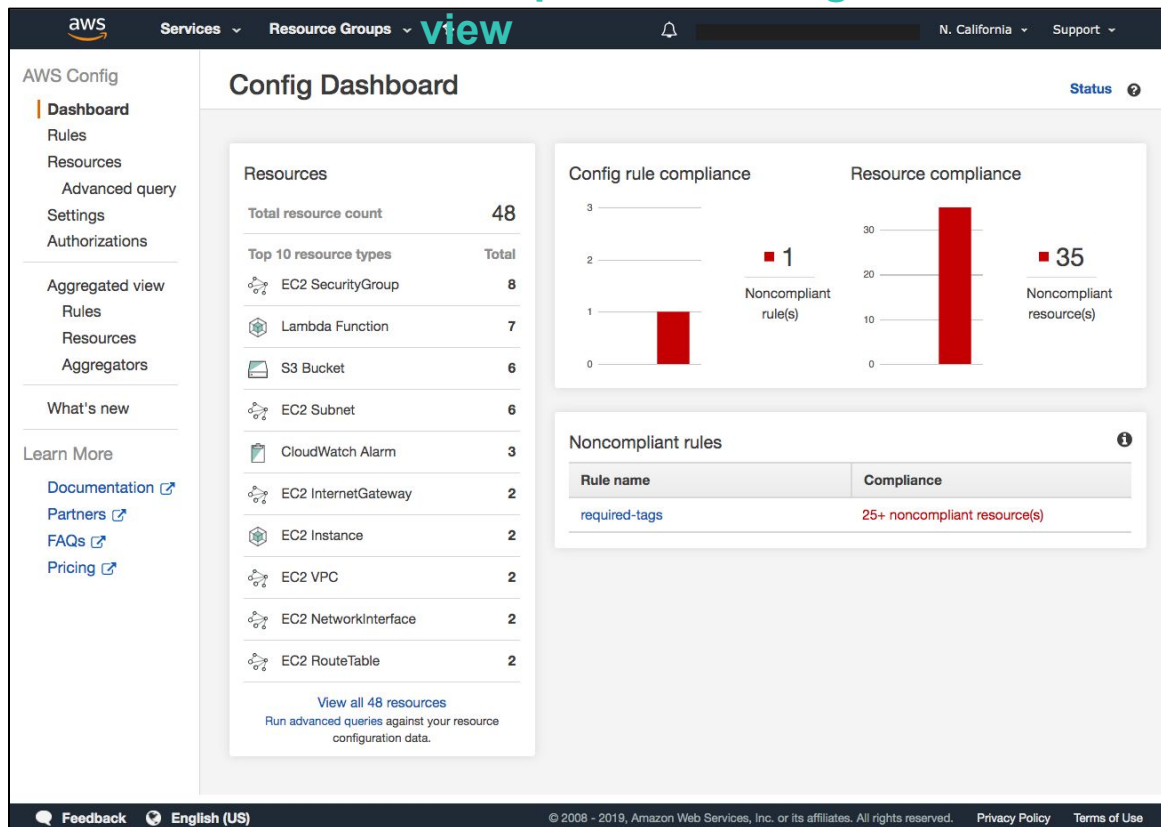


AWS Config



AWS Config

Example AWS Config Dashboard



- **Assess, audit, and evaluate the configurations of AWS resources.**
- Use for continuous monitoring of configurations.
- Automatically evaluate *recorded* configurations versus *desired* configurations.
- Review configuration changes.
- View detailed configuration histories.
- **Simplify compliance auditing and security analysis.**

AWS Artifact



AWS Artifact

- **Is a resource for compliance-related information**
- Provide access to security and compliance reports, and select online agreements
- Can access example downloads:
 - AWS ISO certifications
 - Payment Card Industry (PCI) and Service Organization Control (SOC) reports
- Access AWS Artifact directly from the AWS Management Console
 - Under **Security, Identify & Compliance**, click **Artifact**.

Section 6 key takeaways



- **AWS security compliance programs** provide information about the policies, processes, and controls that are established and operated by AWS.
- **AWS Config** is used to assess, audit, and evaluate the configurations of AWS resources.
- **AWS Artifact** provides access to security and compliance reports.

Section 7: Additional security services and resources

Module 4: AWS Cloud Security

AWS Service Catalog



AWS Service
Catalog

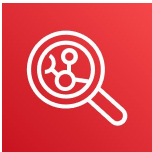
- **Create and manage catalogs of IT services that are approved by your organization**
 - Helps employees find and deploy *approved* IT services
 - An IT service can include one or more AWS resources
 - Example:
 - EC2 instances, storage volumes, databases, and networking components
- **Control AWS service usage by specifying constraints –**
 - Example constraints:
 - The AWS Region where a product can be launched
 - Allowed IP address ranges
- **Centrally manage the IT service lifecycle**
- **Help meet compliance requirements**

Selected additional security services



Amazon
Macie

Proactively **protect personally identifiable information** (PII) and know when it moves.



Amazon
Inspector

Define standards and best practices for your applications and **validate adherence to** these **standards**.



Amazon
GuardDuty

Intelligent **threat detection** and continuous monitoring to protect your AWS accounts and workloads.

Module wrap-up

Module 4: AWS Cloud Security



Module summary

In summary, in this module you learned how to:

- Recognize the shared responsibility model
- Identify the responsibility of the customer and AWS
- Recognize IAM users, groups, and roles
- Describe different types of security credentials in IAM
- Identify the steps to securing a new AWS account
- Explore IAM users and groups
- Recognize how to secure AWS data
- Recognize AWS compliance programs

Complete the knowledge check



Sample exam question

Which of the following is AWS's responsibility under the AWS shared responsibility model?

Choice	Response
A	Configuring third-party applications
B	Maintaining physical hardware
C	Securing application access and data
D	Managing custom Amazon Machine Images (AMIs)

Sample exam question answer

Which of the following is AWS's responsibility under the AWS shared responsibility model?

The correct answer is B.

The keywords in the question are “AWS’s responsibility” and “AWS shared responsibility model”.

Additional resources

- AWS Cloud Security: <https://aws.amazon.com/security/>
- AWS Security Resources:
https://aws.amazon.com/security/security-learning/?cards-top.sort-by=item.additionalFields.sortDate&cards-top.sort-order=desc&awsf.Types=*all
- AWS Security Blog: <https://aws.amazon.com/blogs/security/>
- Security Bulletins :
https://aws.amazon.com/security/security-bulletins/?card-body.sort-by=item.additionalFields.bulletinId&card-body.sort-order=desc&awsf.bulletins-flag=*all&awsf.bulletins-year=*all
- Vulnerability and Penetration testing:
<https://aws.amazon.com/security/penetration-testing/>
- AWS Well-Architected Framework – Security pillar:
<https://d1.awsstatic.com/whitepapers/architecture/AWS-Security-Pillar.pdf>
- AWS documentation - IAM Best Practices:
 <https://docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html>

Thank you

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